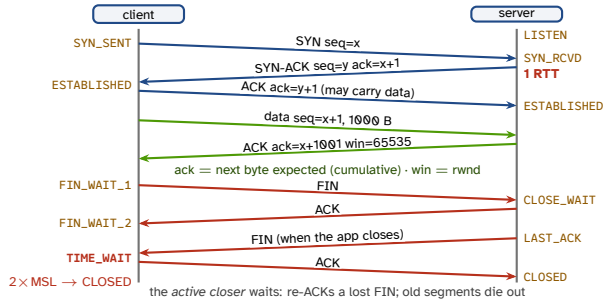
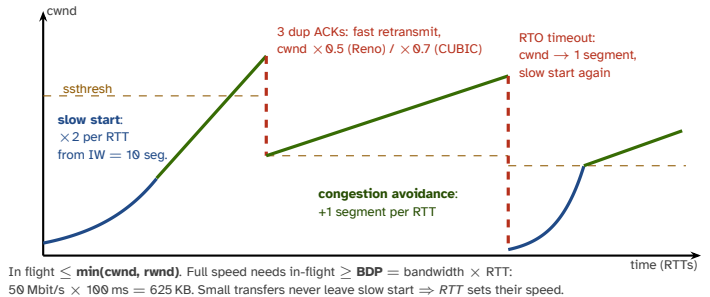


In one line: **TCP** turns lossy IP packets into one **reliable, ordered byte stream**: a 3-way handshake agrees on sequence numbers, ACKs + retransmission recover loss, the **receive window** protects the receiver (flow control), the **congestion window** protects the network. **UDP** is ports + checksum and nothing else. **QUIC** rebuilds TCP's guarantees *per stream* on UDP with TLS 1.3 inside — 1-RTT setup, no cross-stream head-of-line blocking, survives an IP change.

Open, send, close — with the states



Congestion window over time (Reno shape; CUBIC grows as a cubic)



TCP

- Header** (20–60 B): ports · seq (number of the first payload byte) · ack (next byte expected) · flags SYN ACK FIN RST PSH URG ECE CWR · 16-bit window · checksum · options (MSS, window scale $\leq 14 \Rightarrow 1\ GiB$, SACK, timestamps). The 4-tuple (IPs + ports) is the connection.
- Open:** random ISNs (anti-spoofing); SYN and FIN each consume a sequence number; MSS 1460 on a 1500-byte MTU (IPv4). **Close:** each direction separately (half-close) $\rightarrow 4$ segments; the first closer holds **TIME_WAIT** $2 \times MSL$ (RFC MSL 2 min; Linux 60 s). RST aborts at once.
- Loss:** 3 duplicate ACKs $\rightarrow$ fast retransmit; RTO expiry $\rightarrow$ retransmit + slow start from 1 segment. **SACK** reports which ranges arrived.
- Flow control $\neq$ congestion control:** **rwnd** = receiver's free buffer, in every ACK ($0 \rightarrow$ sender stops, probes); **cwnd** = sender's guess of the network. Reno, CUBIC (RFC 9438, Linux's default) are loss-based; **BBR** (Google) models bottleneck bandwidth + min RTT and paces — better on lossy radio and against bufferbloat.
- Nagle** (RFC 896) holds small segments while data is unacked; **delayed ACK** (RFC 1122: $< 0.5\ s$, every 2nd full segment) holds the ACK $\Rightarrow$ write-write-read stalls tens to hundreds of ms. Fix: TCP_NODELAY (noDelay) or one write.
- Head-of-line blocking:** in-order delivery — one lost segment stalls every byte behind it, i.e. every HTTP/2 stream on the connection.

UDP — what you do not get

8-byte header (ports, length, checksum). No handshake, order, retransmission or congestion control; message boundaries kept; oversize datagrams fragment or drop. Used by DNS, QUIC, RTP/WebRTC, games, mDNS. NATs forget idle UDP mappings fast $\rightarrow$ keepalives.

Example — Network.framework knobs

```
import Network
let tcp = NWProtocolTCP.Options()
tcp.noDelay = true // Nagle off: small RPC-style writes
tcp.connectionTimeout = 5 // seconds for the handshake
tcp.enableKeepalive = true; tcp.keepaliveIdle = 30
let conn = NWConnection(host: "api.example.com", port: 443,
    using: NWParameters(tls: .init(), tcp: tcp))
conn.stateUpdateHandler = { state in // preparing/ready/waiting/failed
    if case .waiting(let err) = state { print("no path yet", err) } }
conn.start(queue: .main)
// QUIC (iOS 15): NWParameters(quick: NWProtocolQUIC.Options(alpn: ["h3"]))
// URLSession: h3 when offered (iOS 15); MPTCP: multipathServiceType
```

QUIC — why it exists (RFC 9000/9001/9002, 2021)

- Ossification:** TCP lives in kernels and middleboxes that drop the unfamiliar. QUIC runs in user space over UDP and **encrypts almost all of its header** (even packet numbers), so the network cannot come to depend on it.
- Streams** native, loss per stream. Stream-ID bit 0 = opener (client 0 / server 1), bit 1 = bidi 0 / uni 1. **Packet numbers never reused** (a retransmission is a new packet) $\rightarrow$ unambiguous RTT samples; ACK ranges = built-in SACK.
- Migration:** **connection IDs**, not the 4-tuple; a new IP is checked with PATH_CHALLENGE/PATH_RESPONSE and the connection continues.
- Anti-amplification:** the client pads its first Initial to $\geq 1200\ B$; an unvalidated server sends $\leq 3 \times$ what it received.
- Costs:** more CPU than kernel TCP; UDP blocked or throttled on some networks $\rightarrow$ clients race / fall back to TCP; opaque to middleboxes (by design).

Compare

	TCP	UDP	QUIC
setup	1 RTT (+1 TLS 1.3)	0	1 RTT incl. TLS; 0-RTT resume
delivery	ordered bytes	none	ordered <i>per stream</i>
HOL	whole connection	—	per stream
congestion	kernel: CUBIC, BBR	yours	in app (RFC 9002: NewReno)
identity	4-tuple	—	connection ID $\rightarrow$ migrates
crypto	optional TLS on top	none	mandatory, headers too

Traps

- ! Flow control $\neq$ congestion control: receiver buffer (rwnd) vs network capacity (cwnd).
- ! Thousands of TIME_WAITs on a server: the server closes first. Let clients close, keep connections alive; don't shrink MSL.
- ! Half-open sockets look alive; TCP keepalive defaults to 2 h — use app heartbeats.
- ! More bandwidth rarely speeds a small API call: it is RTT- and slow-start-bound.

Remember

"rwnd: receiver, cwnd: network, TIME_WAIT: who hangs up first; QUIC = TCP per stream."